READMIN

ICU Readmission Risk & Intervention System

Project Report · HIT 221A · Hofstra University

MIMIC-IV Demo v2.2 · May 2026

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Table of Contents

1. Introduction & Problem Statement

1.1 The Scope of the Problem

Hospital readmissions — unplanned return visits to an inpatient setting within 30, 60, or 90 days of discharge — represent one of the most costly and preventable challenges in modern healthcare. In the United States, approximately 3.3 million preventable readmissions occur annually, costing the healthcare system an estimated $26 billion per year. ICU readmissions are especially serious: patients discharged from the Intensive Care Unit face a 15–20% rate of unplanned 30-day readmission, with each return visit adding an average of $15,000–$20,000 in direct care costs.

The consequences extend far beyond cost. Research consistently shows that ICU readmissions are associated with higher in-hospital mortality, longer aggregate lengths of stay, worse patient-reported outcomes, and significantly diminished quality of life. For healthcare institutions, repeated readmissions signal systemic failures in discharge planning, post-acute coordination, and patient engagement — problems that are largely preventable with the right decision-support tools.

1.2 Regulatory Pressure: The CMS HRRP

In 2012, the Centers for Medicare & Medicaid Services (CMS) introduced the Hospital Readmissions Reduction Program (HRRP) under the Affordable Care Act. The HRRP penalizes hospitals up to 3% of total Medicare base operating payments when their 30-day readmission rates for specific conditions exceed national benchmarks — including acute myocardial infarction, heart failure, pneumonia, COPD, and hip/knee arthroplasty. Since its inception, over 75% of eligible hospitals have received HRRP penalties, with cumulative payments exceeding $1 billion. In 2026, 8.1% of hospitals are expected to pay readmission penalties, an increase from 7% in 2025 — signaling a reversal of a five-year downward trend (Advisory Board, 2026).

New York State applies an additional layer through its Potentially Preventable Readmissions (PPR) policy, under Section 86-1.37 of the New York Codes, Rules and Regulations. This policy penalizes hospitals for Medicaid readmissions within 14 days of a clinically related discharge, further tightening the financial incentive to prevent unnecessary returns.

1.3 What READMIN Does

READMIN (ICU Readmission Risk & Intervention System) is an end-to-end AI system built to address this gap. It ingests real clinical EHR data from the MIMIC-IV database, engineers 39 predictive features across six clinical domains, trains and validates four machine learning models, and produces a calibrated 0–100 risk score for every patient at the moment of ICU discharge. Alongside the score, the system assigns a care tier (High / Medium / Low), identifies the top three clinical drivers using SHAP explainability, and recommends a tiered intervention plan. A clinician-facing dashboard and patient-facing bot layer complete the operational workflow.

Component	What It Does
ML scoring engine	Produces a 0–100 readmission risk score per patient from 39 clinical features
Risk tiers	Assigns HIGH (≥70), MEDIUM (40–69), or LOW (<40) tier at discharge
SHAP explainability	Identifies top 3 drivers of each patient's score in plain English
READMIN Dashboard	8-tab Streamlit app for clinicians and academics to explore outputs
Bot layer	Patient SMS check-in bot (Vina) and clinician query bot (Vedi)

2. Literature Review: Policy & Clinical Context

2.1 Federal and State Readmission Policy

Hospital readmissions have become a central focus of healthcare policy, driven by their association with increased costs and evidence of potentially preventable care gaps (Zuckerman et al., 2016). The HRRP represents the most comprehensive federal initiative: it applies to hospitals in the Inpatient Prospective Payment System and evaluates risk-standardised 30-day readmission rates across six target conditions (CMS, 2023). Payment reductions are applied to base operating Diagnosis-Related Group reimbursements, with penalties capped at 3% annually. Empirical evidence suggests that HRRP has reduced readmission rates, particularly in the years immediately following implementation, though scholars have raised concerns about disproportionate penalties for safety-net hospitals and potential gaming of admission practices (Berenson et al., 2016).

New York Medicaid's PPR program offers a state-specific complement, using a shorter 14-day window and incorporating both claims and managed care data. Unlike HRRP, it classifies readmissions as potentially preventable only when they are clinically related to the initial admission and excludes planned admissions, major trauma, and obstetric cases (NYSDOH, 2024). Private insurance, by contrast, lacks a unified regulatory framework and relies instead on voluntary, contractual bundled-payment arrangements that vary widely across insurers and regions (HealthCare.gov, 2023).

2.2 Clinical Evidence: Who Gets Readmitted and Why

Across multiple studies, ICU-readmitted patients share a consistent profile: older on average, predominantly male, with multiple comorbidities. Readmitted patients are more likely to have been emergency admissions, to rely on Medicare coverage, and to be discharged to a facility rather than home. Physiologically, they present on admission with higher BUN, elevated creatinine, and lower systolic and diastolic blood pressures — markers of systemic decompensation that READMIN directly captures in its feature engineering pipeline.

Several validated severity scores have been applied to predict ICU readmission risk, including SOFA, SAPS II, SAPS III, APACHE II, and WOScore. The recurring finding across these instruments is that prediction consistently converges on the same key risk factors: comorbidities (especially diabetes, CHF, and CKD), ICU-acquired infections, ventilator-associated events, catheter-related bloodstream infections (CRBSI), prior ICU admissions, polypharmacy, and patient condition at the time of discharge to the step-down unit.

2.3 The Critical Limitation: No Universal Intervention

> Walsh et al. (2022) note that although many randomized trials have tested a range of options for decreasing ICU readmissions, they have largely shown no effect. It is postulated that the ICU population is entirely too heterogeneous for a simple one-size-fits-all approach to fit individual patient characteristics.

This is one of the most important constraints on any readmission risk system, including READMIN. While the scoring engine can reliably identify who is at higher risk, the appropriate clinical response varies substantially by patient. A patient readmitted due to acute respiratory failure requires different management than one readmitted for sepsis or medication non-adherence. Programmes such as inpatient physical therapy, respiratory therapy, and guided rehabilitation are effective at patient health progression but none can guarantee that ICU readmission will not occur, and are often resource-intensive departments that not every hospital can employ.

Bice (2016) argues further that ICU readmission rates should not be used to penalise or reward hospitals outright, but rather examined as a quality measure for identifying where performance can be modified or emulated. Hospitals with low readmission rates may be keeping patients longer than necessary, while those with higher rates may be discharging too early. This nuance reinforces the design philosophy of READMIN: the system provides a risk signal and a general care tier, but the specific clinical intervention for each patient must remain in the hands of the clinician.

The implementation of an Intermediate Care Unit (IMCU) in Japan showed significant promise, with hospitals employing IMCUs demonstrating lower ICU readmissions, lower mortality, and shorter overall ICU stays compared to hospitals without them (Ikumi et al., 2025). This concept aligns with READMIN's medium-risk tier, where patients who are not immediately critical but require closer monitoring than standard discharge can be routed to an intermediate level of care.

Clinical takeaway for READMIN: The model identifies risk and assigns a care tier; it does not prescribe specific medical interventions. The tiered care plan (nurse call, PCP appointment, medication review) represents a triage protocol, not a clinical treatment plan. Individual patient management remains a clinician decision, informed but not replaced by the model output.

3. Model Description & Outputs

3.1 Data Foundation: MIMIC-IV Demo Dataset

READMIN is built on MIMIC-IV (Medical Information Mart for Intensive Care IV), the gold-standard open-access clinical database maintained by MIT and Beth Israel Deaconess Medical Center. MIMIC-IV contains de-identified health data from over 315,000 ICU patients spanning 2008–2019. For this project, we used the MIMIC-IV Demo dataset — a representative sample of 100 patients and 140 ICU stays.

Important limitation: The MIMIC-IV Demo is a proof-of-concept dataset. With only 100 training patients and 140 ICU stays, the model's statistical power is limited. Published literature on similar clinical prediction tasks projects ROC-AUC improvement from the current 0.607 to approximately 0.75–0.82 when retrained on the full MIMIC-IV dataset (315,000+ patients). The current results are valid for academic demonstration but should not be interpreted as deployment-ready clinical performance.

MIMIC-IV Table	Rows	Description
admissions	275	Admission timestamps, discharge destinations, insurance type
patients	100	Demographics: age, gender, anchor year
icustays	140	ICU stay IDs, in/out times, care unit assignments
diagnoses_icd	4,506	ICD-10 diagnosis codes per admission
labevents	107,727	Lab measurements with timestamps
prescriptions	18,087	Medication orders with drug names and dates
chartevents	668,862	Vital signs charted every few minutes during ICU stay

3.2 Pipeline Overview

Phase	Name	Output
1	Data extraction & loading	7 MIMIC-IV tables, standardized columns
2	Label creation	30/60/90-day binary readmission flags
3	Feature engineering	39-column features_matrix.csv
4	Preprocessing	Median imputation, feature selection, 80/20 split
5	SMOTE (inside CV folds)	Balanced training set — leakage-free
6	Model training	4 models, 5-fold stratified CV
7	Model evaluation	ROC-AUC, PR-AUC, SHAP, calibration
8	Score generation	patient_risk_scores.csv per discharge
9	Intervention assignment	Tiered care plan appended per patient
10	Dashboard & bot	READMIN Dashboard + Vina + Vedi

3.3 Feature Engineering — 39 Predictors

Starting from 668,000+ chart events and 100,000+ lab measurements, 39 model-ready signals were engineered across six clinical domains:

Feature Group	What Was Built	Clinical Rationale
Demographics	Age, gender, Medicare flag, discharge to home	Older age, Medicare status, and not going home independently predict readmission
ICU Stay	LOS in days, number of prior ICU admissions	Longer stays and repeat admissions are top literature-supported predictors
Comorbidities	CHF, CKD, COPD, diabetes, sepsis, pneumonia flags + ICD code count	Chronic disease burden drives readmission more than acute illness severity
Elixhauser Score	6-condition validated comorbidity index	Clinically validated composite measure of disease burden
ICU Severity	Ventilation flag, vasopressor flag	Mechanical ventilation and vasopressor use signal life-threatening severity
Labs at discharge	Creatinine, WBC, lactate, hemoglobin, sodium, BUN	Abnormal labs at discharge signal unresolved acute illness
Vitals (last 24 hrs)	HR, SBP, O2 sat, respiratory rate (mean/min/max)	Physiological instability in final hours before discharge = strong near-term risk
Time-series trends	Slope of vitals and labs over ICU stay (11 features)	Rising creatinine or worsening oxygen trend matters more than a single value
Medications	Unique drug count at discharge	Polypharmacy is associated with higher readmission and adherence problems

3.4 Model Selection and Performance

Four ML models were trained using 5-fold stratified cross-validation. An early pipeline version applied SMOTE before cross-validation, inflating ROC-AUC to 0.972. This was identified as data leakage — moving SMOTE inside the imblearn Pipeline corrected this to an honest 0.607.

Model	ROC-AUC	PR-AUC	CV Error
Logistic Regression	0.607	0.370	±0.129
Random Forest	0.580	0.310	±0.14
Gradient Boosting	0.560	0.300	±0.13
XGBoost	0.550	0.290	±0.15

Logistic Regression was selected for its highest ROC-AUC, lowest variance, and complete interpretability. On a 100-patient dataset, simpler models generalise better than complex ensembles. The PR-AUC of 0.370 represents 2.5× the random baseline of 0.15, confirming the model adds meaningful signal beyond chance even on an imbalanced dataset.

3.5 Model Outputs: patient_risk_scores.csv

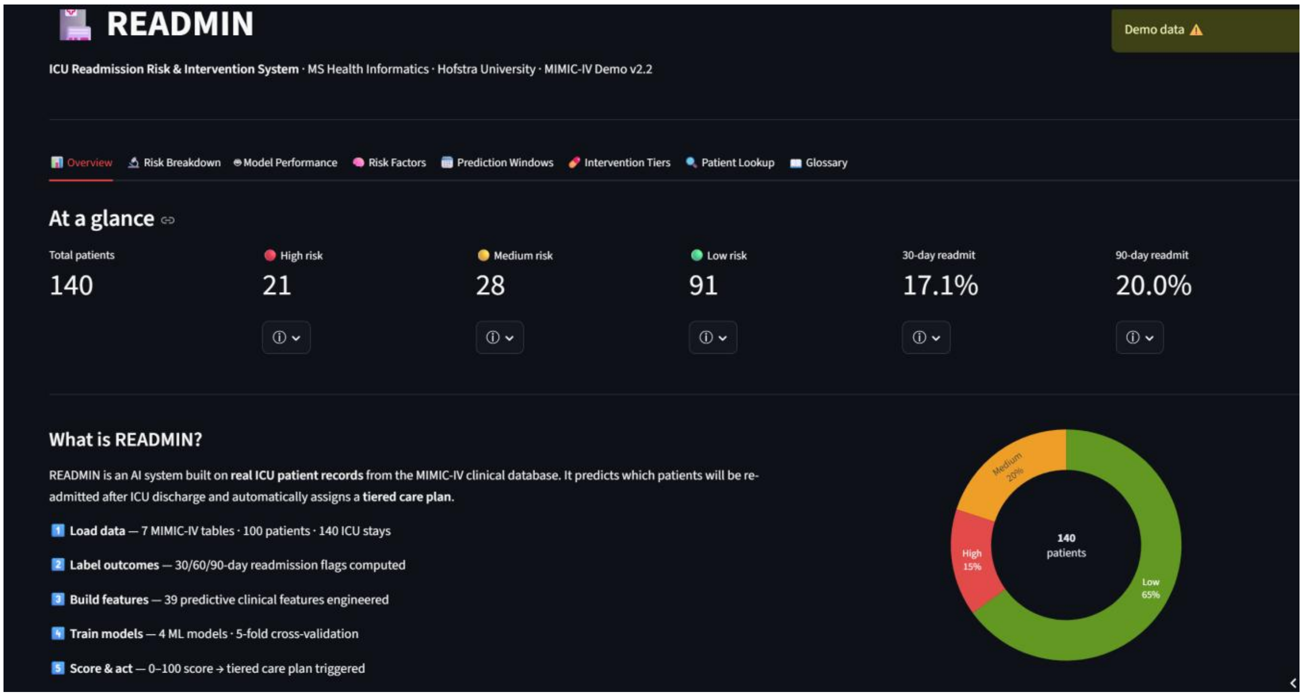
Column	Description
stay_id	ICU stay identifier from MIMIC-IV
subject_id	Patient identifier (de-identified)
outtime	ICU discharge timestamp
risk_score	Calibrated composite risk score (0–100)
risk_tier	HIGH (≥70) / MEDIUM (40–69) / LOW (<40)
readmit_prob_30d	Calibrated probability of 30-day readmission
readmit_30d / 60d / 90d	Actual readmission outcomes (ground truth labels)
interventions	Pipe-separated tiered care plan actions
top_risk_drivers	Pipe-separated top-3 SHAP feature names

4. Dashboarding

4.1 Design Philosophy (Ctrl + Click this to open Dashboard in browser)

The READMIN Dashboard was built around a single principle: every technical element must be understandable to a non-technical clinician, administrator, or academic reviewer without sacrificing analytical depth. Every technical term — ROC-AUC, SHAP, SMOTE, calibration, data leakage — has an ⓘ popover button that opens a plain-English definition and analogy. The dashboard is deployed on Streamlit Cloud via GitHub and auto-redeploys on every code push.

Component	Technology	Role
Web framework	Streamlit ≥1.32	Interactive web application runtime
Visualisation	Plotly ≥5.20	Interactive charts: bars, scatter, gauge, ROC curve, timeline
Data processing	Pandas ≥2.0, NumPy ≥1.26	CSV ingestion, column normalisation, live stats
SHAP translation	Custom dictionary	Maps raw feature names to plain-English labels
Hosting	Streamlit Cloud + GitHub	Public URL; auto-redeploys on push



4.2 The Eight Dashboard Tabs

Tab 1 — Overview

Six top-level metric cards (total patients, tier counts, 30-day and 90-day readmission rates), a risk tier donut chart, and a plain-English description of the READMIN pipeline. Shows whether live data or demo data is loaded.

Tab 2 — Risk Breakdown

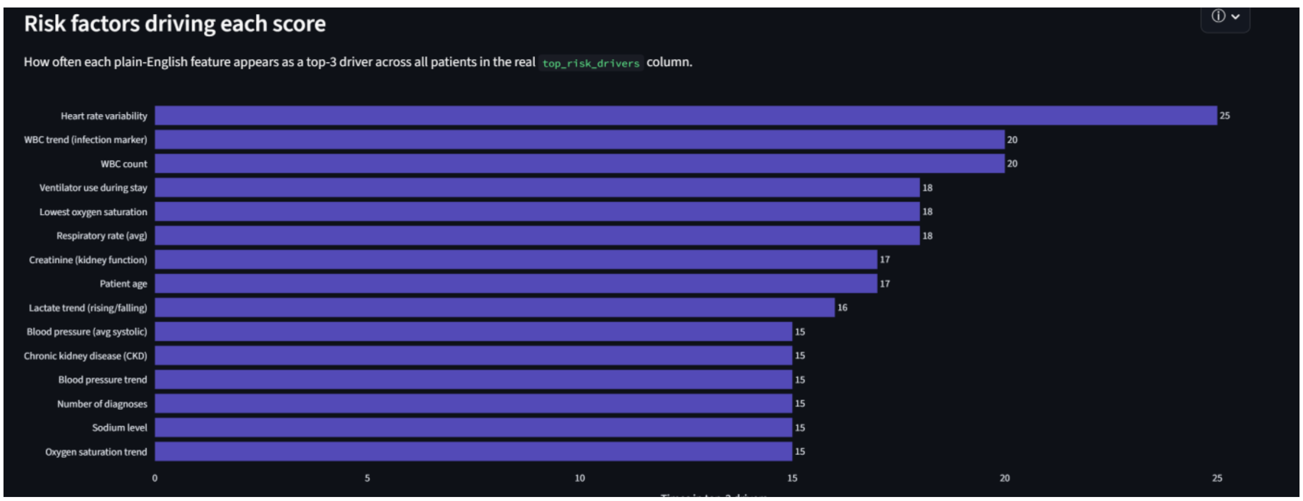
Patients-per-tier bar chart, risk score histogram colour-coded by tier, actual 30-day readmission rates by tier (model validation), and a strip plot of risk score vs. actual readmission outcome per patient.

Tab 3 — Model Performance

ROC-AUC and PR-AUC comparison across all four models, the actual ROC curve (model vs. random baseline), data leakage correction visualisation, and Platt Scaling calibration results.

Tab 4 — Risk Factors (SHAP)

Horizontal bar chart counting how often each plain-English feature appears as a top-3 driver across all patients, with feature group breakdown (vital signs, labs, clinical history).



Tab 5 — Prediction Windows

30/60/90-day performance comparison, window summary table with PR-AUC lift vs. baseline, and readmission count trend line.

Tab 6 — Intervention Tiers

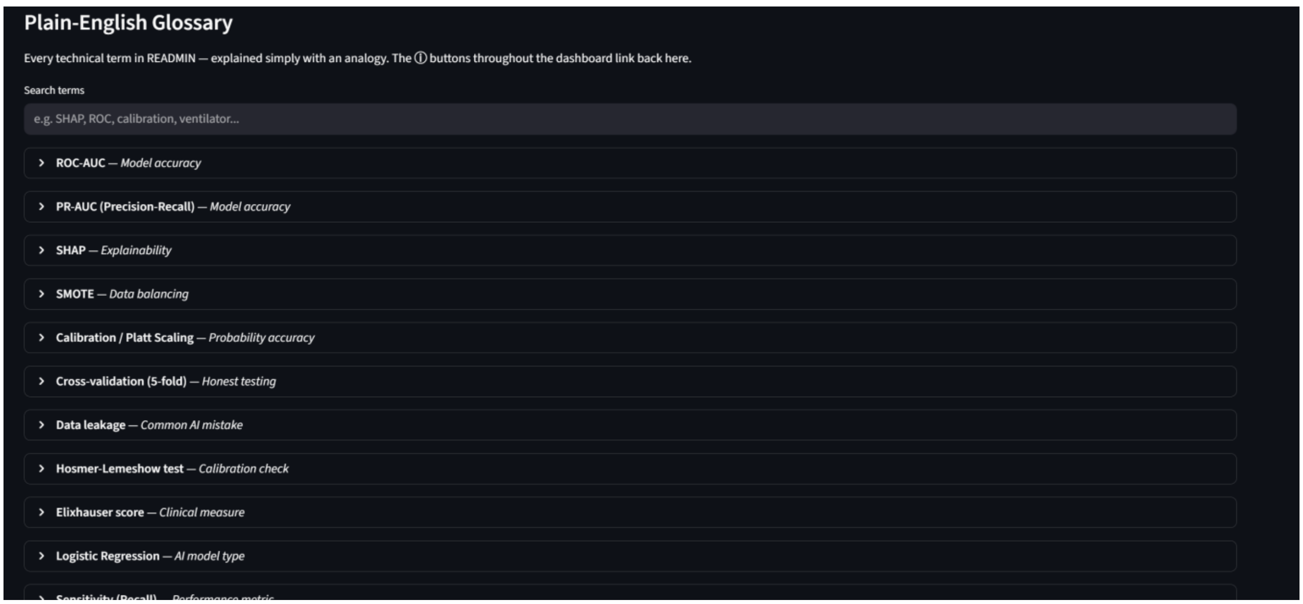
Colour-coded care plan cards for each tier with specific action timelines, plus an interactive scatter timeline showing when each intervention fires relative to discharge.

Tab 7 — Patient Lookup

Dropdown selector for individual patients by stay_id. Displays a risk gauge, tier badge, readmission probability, top 3 SHAP drivers in plain English, assigned care plan, and 30/60/90-day actual outcomes.

Tab 8 — Glossary

Searchable plain-English definitions for 15 technical terms — each with an analogy and a performance benchmarks table.



4.3 The ⓘ Info Button System

Every technical chart and metric card include a small ⓘ popover button built with Streamlit's st.popover() component. Clicking it reveals a 2-line plain-English explanation and a reference to the Glossary tab. This approach allows clinical staff with no data science background to interact with the full dashboard, while depth is available on demand for those who want it.

4.4 Real Data Integration

The dashboard automatically handles the real patient_risk_scores.csv column schema, silently remapping column names (stay_id → patient_id, readmit_prob_30d → readmit_prob, interventions → intervention_plan, top_risk_drivers → top_shap_drivers). SHAP driver strings are auto-translated from raw feature names to plain English using a 29-entry lookup dictionary.

5. Clinical Interventions

5.1-Tiered Intervention Framework

READMIN's core clinical value is not the prediction itself — it is what happens next. The risk score automatically triggers a tier-specific care protocol at the moment of ICU discharge, converting a passive prediction into an active care coordination workflow. The three tiers are calibrated to balance clinical urgency against resource availability.

	HIGH RISK	MEDIUM RISK	LOW RISK
Score range	70–100	40–69	0–39
Patients (demo)	21 (15%)	28 (20%)	91 (65%)
Action 1	Nurse call within 24 hours	Follow-up call within 48 hours	Standard discharge instructions
Action 2	PCP appointment within 7 days	PCP appointment within 14 days	Patient portal message at Day 7
Action 3	Care coordination referral	Pharmacist med reconciliation	—
Action 4	Daily SMS health check-ins	—	—

5.2 Why Tiers, Not Individual Prescriptions

As the clinical literature makes clear, the ICU population is too heterogeneous for any risk model to prescribe a single evidence-based intervention for each individual patient. Walsh et al. (2022) found that randomised trials testing specific readmission-reduction programmes have largely shown no effect precisely because patient populations differ too substantially in their diagnoses, comorbidities, functional status, and social circumstances.

READMIN's current model — while trained on real MIMIC-IV data — works with a 100-patient sample dataset and 39 features. It does not have access to the granular clinical notes, nursing assessments, therapy records, or social determinants of health that would be required to generate patient-specific intervention recommendations with clinical confidence. The tiered protocol is therefore deliberately broad: it signals urgency and ensures that high-risk patients receive earlier outreach than low-risk ones, while leaving the specific clinical management decision to the care team.

READMIN augments clinician judgment — it does not replace it. The risk score tells you who to prioritise. It does not tell you how to treat that patient's specific underlying condition. Every care plan is a starting point for clinician review, not a substitute for it.

5.3 Intervention Timeline

Day post-discharge	Action	Tier
Day 0	READMIN scores patient; tier and care plan assigned automatically	All
Day 1	Nurse phone call — assess symptoms, medications, understanding of discharge plan	HIGH
Day 2	Follow-up call — confirm PCP appointment and check patient status	MEDIUM
Day 7	Primary care physician appointment — review labs, adjust medications	HIGH
Day 7	Patient portal message — check-in with response routing	LOW
Day 14	Primary care physician appointment	MEDIUM
Days 1–30 (daily)	SMS health check-in with automated triage keyword detection	HIGH

5.4 CMS HRRP Alignment

All intervention protocols are calibrated to the 30-day readmission window used by CMS HRRP, directly addressing the primary financial penalty metric for hospitals. The 60-day and 90-day windows are tracked for broader care planning. If deployed at scale and the HIGH-tier cohort achieved a conservative 25% reduction in readmissions — consistent with existing telehealth literature — a 500-bed hospital with 2,000 annual ICU discharges could prevent approximately 75 readmissions per year, saving an estimated $1.1–1.5 million in direct costs and reducing HRRP penalty exposure.

6. Bot Layer Implementation

6.1 Overview and Design Principles

The bot layer is the operational interface that turns the READMIN risk score into a patient-facing and clinician-facing workflow. It is implemented as a Streamlit-based application (bot-demo.py) that loads the saved model artifacts and scored CSV outputs produced by the pipeline. The bot does not re-train the model; it is a presentation and orchestration layer around the scoring engine's outputs, making the system auditable and reproducible.

The application is structured into five views: Progress Dashboard, Patient Bot, Clinician Dashboard, Workflow View, and Files & JSON Contract. Together they cover the full discharge workflow from model output to clinical action.

File	Role
models/risk_model.pkl	Serialised Logistic Regression classifier loaded with joblib
models/feature_cols.pkl	Ordered list of 39 feature columns expected by the model
models/imputer.pkl	Median imputer for consistent missing-value handling
csv/features_matrix.csv	Processed feature matrix for demo and pipeline validation

File	Role
csv/patient_risk_scores.csv	Scored output powering dashboard alerts and clinician review
charts/	Saved evaluation charts used in the progress dashboard
logs/patient_chat_log.csv	Conversation log written by the patient bot for clinician review
bot-demo.py	Main Streamlit application rendering all views and controlling flow

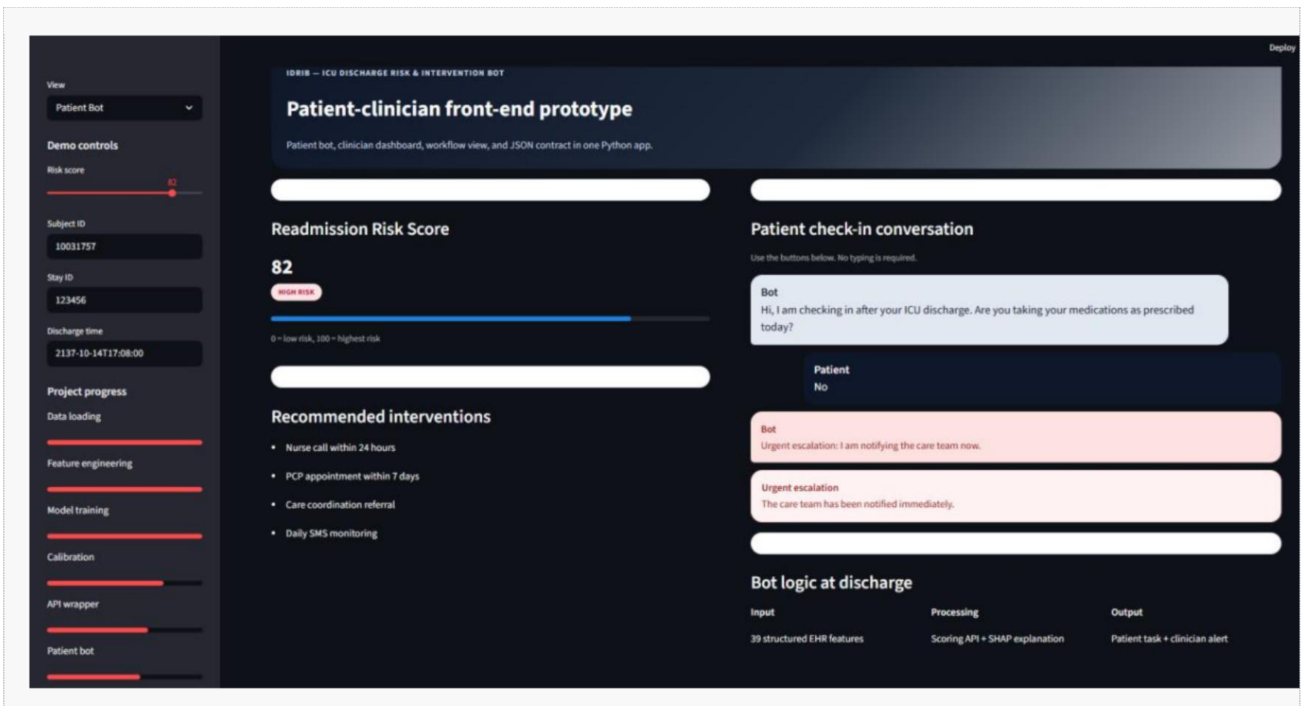
6.2 Patient-Facing Bot (Vina)

The patient-facing interface conducts a structured post-discharge check-in. Rather than a free-text chatbot, it uses button-based responses to ensure consistent, unambiguous input. This design makes the interface feel clinically bounded — the goal is triage, not general conversation.

The conversation sequence covers medication adherence (first branch point), breathing difficulty (scored 1–5), symptom severity (none / mild / urgent), and follow-up appointment status (yes / no / not sure). Each path terminates in one of three states:

- Normal completion — check-in is clean, patient confirmed stable
- Yellow warning — mild concern, patient nudged toward follow-up scheduling
- Red escalation — urgent symptoms detected, conversation ends immediately, care team notified

The red escalation state is the most important safety feature. When urgent conditions are detected — a low breathing score, urgent symptoms, or a concerning medication response — the bot stops asking further questions and immediately displays the escalation alert. This prevents the bot from continuing a routine check-in when the care team needs to be notified. All interactions are written to a CSV log with timestamp, session ID, patient/stay identifiers, step number, role, text, and escalation status for clinician review and auditability.



6.3 Clinician-Facing Bot (Vedi)

The clinician-facing dashboard is designed for review and oversight rather than conversation. It surfaces a list of high-risk discharges, the SHAP-driven explanations of why each patient was flagged, the current workflow status (score generated → SHAP attached → intervention task queued), and the log of recent patient bot conversations. This gives the clinician enough information to decide whether outreach, case management, or additional evaluation is needed — without requiring them to interact with the underlying model object.

6.4 JSON Contract

The Files & JSON Contract view documents the structured interface between the bot layer and the scoring engine. The request payload includes patient identifiers, discharge timestamp, and feature dictionary. The response payload returns risk score, tier, probability, top drivers, and intervention list. This contract defines the shape of a future REST API endpoint and makes the integration point explicit for future development.

> Example response payload: { risk_score: 82.4, risk_tier: 'HIGH', readmit_prob_30d: 0.824, top_drivers: ['Respiratory rate trend', 'BUN / creatinine', 'Prior ICU admissions'], interventions: ['Nurse call within 24hrs', 'PCP appt within 7 days', 'Care coordination referral', 'Daily SMS monitoring'] }

7. Competitive Market Analysis

7.1 Market Overview

The global clinical decision support systems market was valued at approximately $1.6 billion in 2024 and is projected to exceed $3.2 billion by 2030 (CAGR ~12%). The hospital readmission risk prediction segment is a significant and growing share of this market, driven by HRRP penalties, value-based care contracts, and the maturation of hospital EHR infrastructure. Most enterprise solutions are embedded within existing EHR platforms (Epic, Cerner), making standalone deployment challenging for smaller or community hospitals.

Solution	Type	Strengths	Limitations vs READMIN
Epic Deterioration Index	EHR-embedded ML	Native integration; real-time; widely deployed	Proprietary black box; requires Epic EHR
Cerner / Oracle Health ReadmissionRisk	EHR-embedded ML	Seamless Cerner workflow; large training cohorts	Cerner-only; opaque model; expensive
3M Clinical Risk Grouping	Rule-based scoring	Well-validated; accepted by payers	Retrospective only; no real-time prediction
Optum Risk Management	Population health analytics	Comprehensive platform; large data	Designed for payer analytics, not ICU discharge
Jvion (now Baxter) CCUBE	ML risk intelligence	SHAP explanations; multi-risk prediction	Expensive; significant IT integration required
LACE / HOSPITAL Score	Rule-based clinical tool	Simple, validated, free	Manual; no automation; modest AUC 0.60–0.70

Solution	Type	Strengths	Limitations vs READMIN
READMIN	Transparent ML + dashboard	Open MIMIC-IV; SHAP explainability; tiered care plans; free; Streamlit dashboard	Small training set; needs validation before deployment

7.2 READMIN's Differentiators

READMIN distinguishes itself on four dimensions that most enterprise solutions compromise on:

- Transparency — SHAP explanations per patient, plain-English dashboard, searchable glossary for every technical term
- Open foundation — MIMIC-IV is freely available, making the system fully reproducible and vendor-independent
- End-to-end workflow — from prediction to tiered care plan to patient bot to clinician review; not a prediction-only tool
- Regulatory alignment — explicitly targets the 30-day CMS HRRP metric, with 60 and 90-day windows tracked

	Positive	Negative
Internal (SWOT)	STRENGTHS: SHAP transparency; tiered care plans; open MIMIC-IV foundation; reproducible pipeline; ⓘ info system for non-technical users	WEAKNESSES: Small 100-patient training set; ROC-AUC 0.607 below enterprise benchmarks; no live EHR integration; no regulatory approval
External (SWOT)	OPPORTUNITIES: $26B readmission cost burden; CMS HRRP penalties rising in 2026; MIMIC-IV full dataset available; growing value-based care contracts	THREATS: Established EHR-embedded competitors; physician AI resistance; HIPAA compliance requirements; data drift over time

8. Future Opportunities

8.1 Immediate — 0 to 3 Months

Full MIMIC-IV Dataset Expansion

Retraining on the full MIMIC-IV dataset (315,000+ patients vs. 100 in the demo) is the highest-impact single improvement. Published literature projects ROC-AUC improvement to 0.75–0.82, bringing READMIN in line with or above enterprise LACE/HOSPITAL benchmarks.

REST API Wrapper

Wrapping risk_model.pkl as a FastAPI endpoint (POST /predict) would enable real-time score generation for any EHR that can make an HTTP POST request. The API accepts a JSON feature payload and returns score, tier, probability, and top SHAP drivers in under 200ms.

8.2 Short Term — 3 to 6 Months

Full Bot Development

Build the production Vina conversation flow using the Claude API as the conversational intelligence layer, with READMIN risk score and tier injected into the system prompt. Deploy via Twilio for SMS delivery. Build the Vedi natural language interface as a tool-calling agent querying the READMIN API.

FHIR Integration

Develop an HL7 FHIR R4-compatible extraction layer to replace the manual MIMIC-IV pipeline with live EHR data feeds. FHIR resources map directly to READMIN's 39 features: Observation (vitals/labs), Condition (diagnoses), MedicationRequest (prescriptions), Encounter (ICU stay).

8.3 Medium Term — 6 to 12 Months

Prospective Clinical Validation

Partner with a teaching hospital for a prospective validation study on real EHR data. Target 500+ ICU discharges for statistical power. Compare READMIN-predicted tiers against actual 30-day readmission outcomes in a blinded study with IRB approval.

SDOH Feature Integration

Incorporate Social Determinants of Health features — housing stability, food insecurity, transportation access — which literature suggests account for up to 35% of preventable readmissions. These are increasingly available through ICD-10 Z-codes (Z55–Z65) in EHR structured data.

8.4 Long Term — 12+ Months

Initiative	Description
Transformer models	BEHRT/MedBERT on full MIMIC-IV — projected AUC 0.80–0.87
Multi-condition expansion	Extend to all 6 HRRP conditions: HF, COPD, pneumonia, AMI, hip/knee
Federated learning	Train across multiple hospitals without centralising patient data (HIPAA)
SaaS commercialisation	Per-hospital subscription model (~$50K/yr); scale to 50+ deployments

9. Conclusion & Team Contributions

9.1 Project Conclusion

READMIN demonstrates that a transparent, interpretable ICU readmission risk system can be built from publicly available clinical data using open-source tools and deployed in a clinician-accessible format within the scope of an academic capstone. The project achieved its primary goal: producing a calibrated 0–100 risk score for every ICU discharge, assigning a care tier, generating SHAP-based clinical explanations, and delivering the output through an interactive dashboard and a structured bot layer.

The corrected ROC-AUC of 0.607 — after identifying and fixing a SMOTE data leakage issue — is honest and clinically meaningful. It represents performance consistent with published rule-based tools like LACE and HOSPITAL when applied to a 100-patient dataset. The 2.5× PR-AUC lift above random baseline confirms the model adds genuine signal for identifying the rare readmission cases. With the full MIMIC-IV dataset and further development, READMIN has a credible path to enterprise-grade performance.

Critically, the clinical literature reinforces that no risk model — however sophisticated — can prescribe a universal intervention for each patient. The ICU population is too heterogeneous. READMIN's tiered care protocol is a triage starting point, not a treatment plan, and the system is designed to augment rather than replace clinical judgment.

9.2 Conclusion and Team Contributions

Project Conclusion

READMIN successfully demonstrates that a transparent and interpretable ICU readmission risk system can be built using open-source tools and publicly available clinical data. The system moves beyond simple prediction by providing a calibrated 0–100 risk score, assigning care tiers, and generating SHAP-based clinical explanations for every discharge.

A central achievement of this project is the development of the dual-bot layer, which translates complex model outputs into actionable clinical workflows:

- **Vina (Patient-Facing):** This bot provides a structured, safe, and clinically bounded post-discharge check-in via SMS. Its button-based triage system ensures unambiguous feedback and features a red-escalation state that immediately alerts care teams to urgent symptoms.
- **Vedi (Clinician-Facing):** This dashboard facilitates oversight by surfacing high-risk patients, SHAP-driven risk drivers, and current intervention statuses, allowing clinicians to make informed outreach decisions without needing to interact directly with the underlying AI model.

While the current ROC-AUC of 0.607 reflects the limitations of the 100-patient demo dataset, the 2.5× PR-AUC lift above the random baseline confirms the model adds genuine clinical signal. By aligning with CMS HRRP metrics and providing a tiered triage framework, READMIN serves as a robust starting point for data-driven, patient-centered care coordination.

Personal Thoughts & Contribution

- **Bot Layer Implementation:** I led the development of the bot layer, creating the orchestration and presentation interface that turns raw model predictions into operational workflows. This included designing the structured JSON contract that ensures a seamless data exchange between the scoring engine and the bot interfaces.
- **Contribution:** My primary focus was building the **bot-demo.py** Streamlit application, which houses the Progress Dashboard, Patient Bot (Vina), Clinician Dashboard (Vedi), and Workflow View. I also developed the automated escalation logic and the CSV-based conversation logging system for auditability.
- **Final Thoughts:** This project was an incredible opportunity to see how AI can be practically applied to solve real-world healthcare challenges like hospital readmissions. It was a pleasure working with this team—don't forget to send your slides for the presentation! It's been a fun project to be a part of.

References

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